

ASSIGNMENT-24.4

NAME: G.MANI SHARAN

ROLL.NO:2303A52148

BATCH:41

Task 1 – Rapid App Prototyping with AI

Task:

Create a quick mobile app prototype using AI in Replit.

Prompt used:

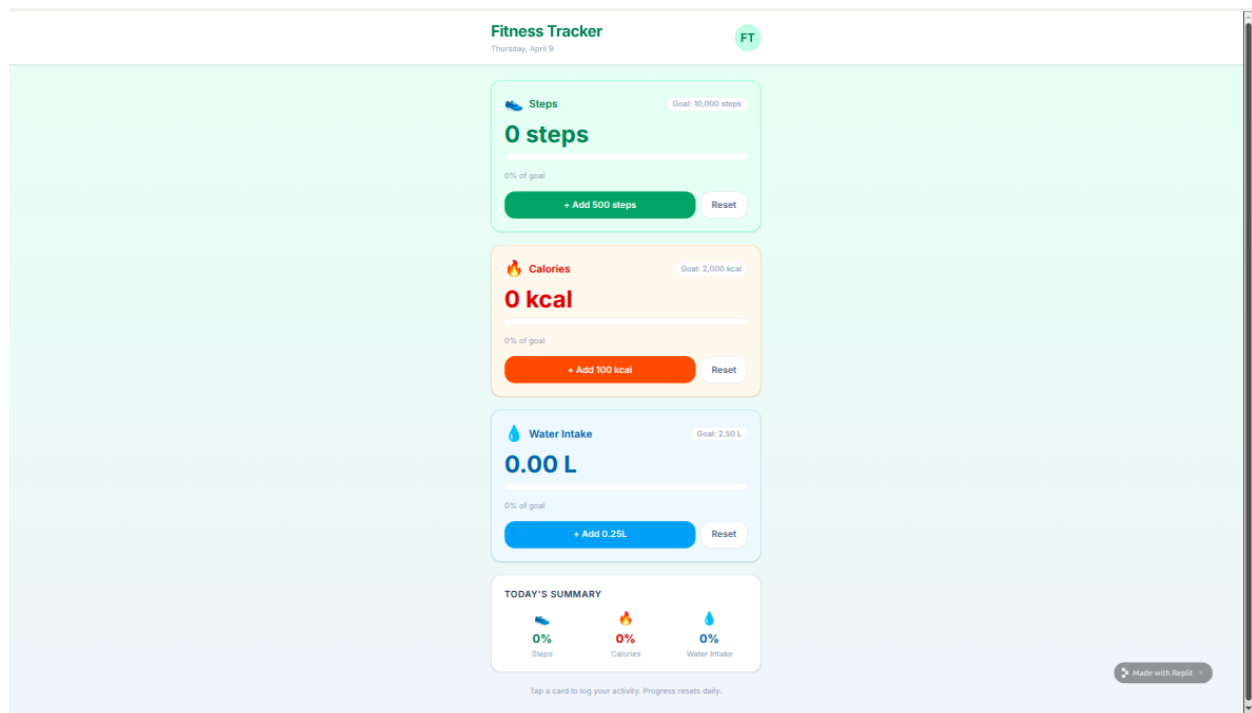
Create a simple mobile-friendly React app UI for a Fitness Tracker.

Include:

- App title "Fitness Tracker"
- Sections: Steps, Calories, Water Intake
- Add placeholder values (e.g., 0 steps, 0 calories, 0L water)
- Use clean layout and basic CSS
- Make it responsive for mobile screens

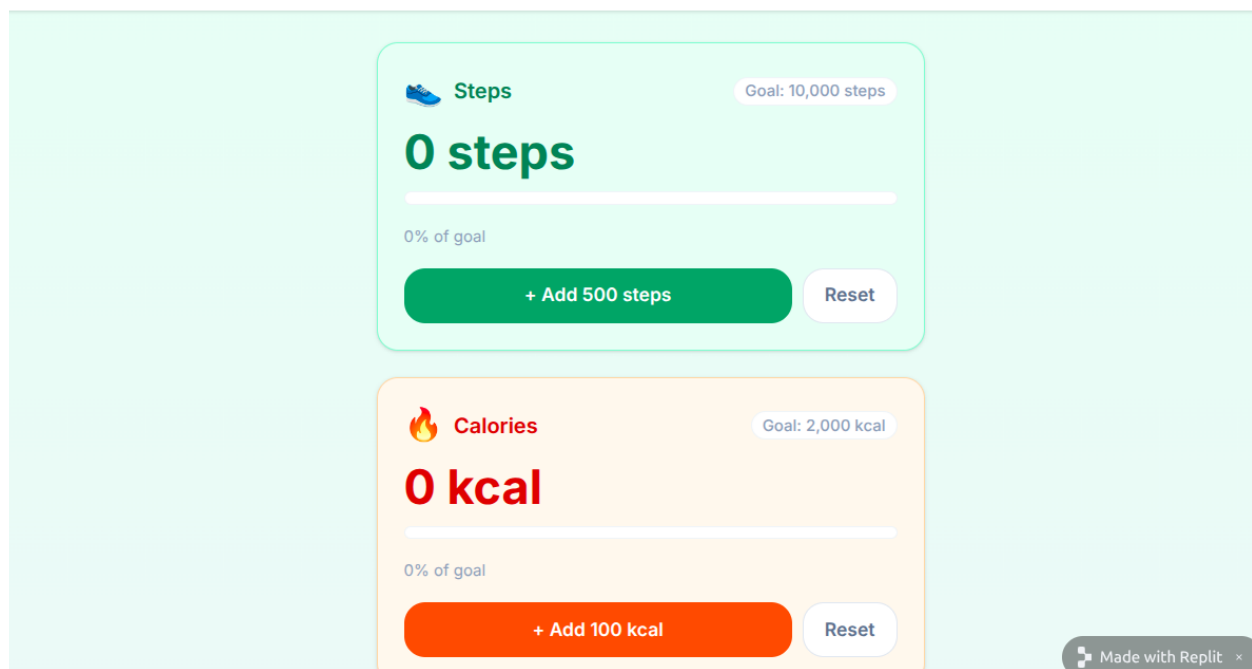
Output:

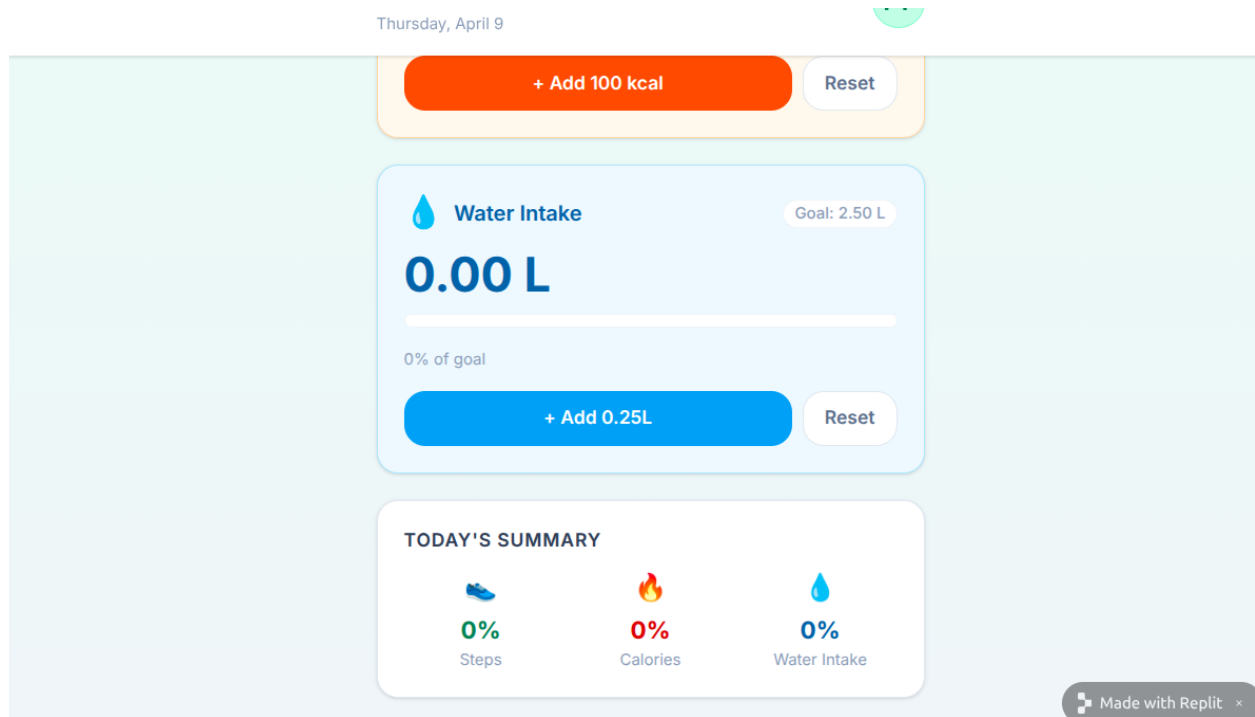
A simple mobile app interface



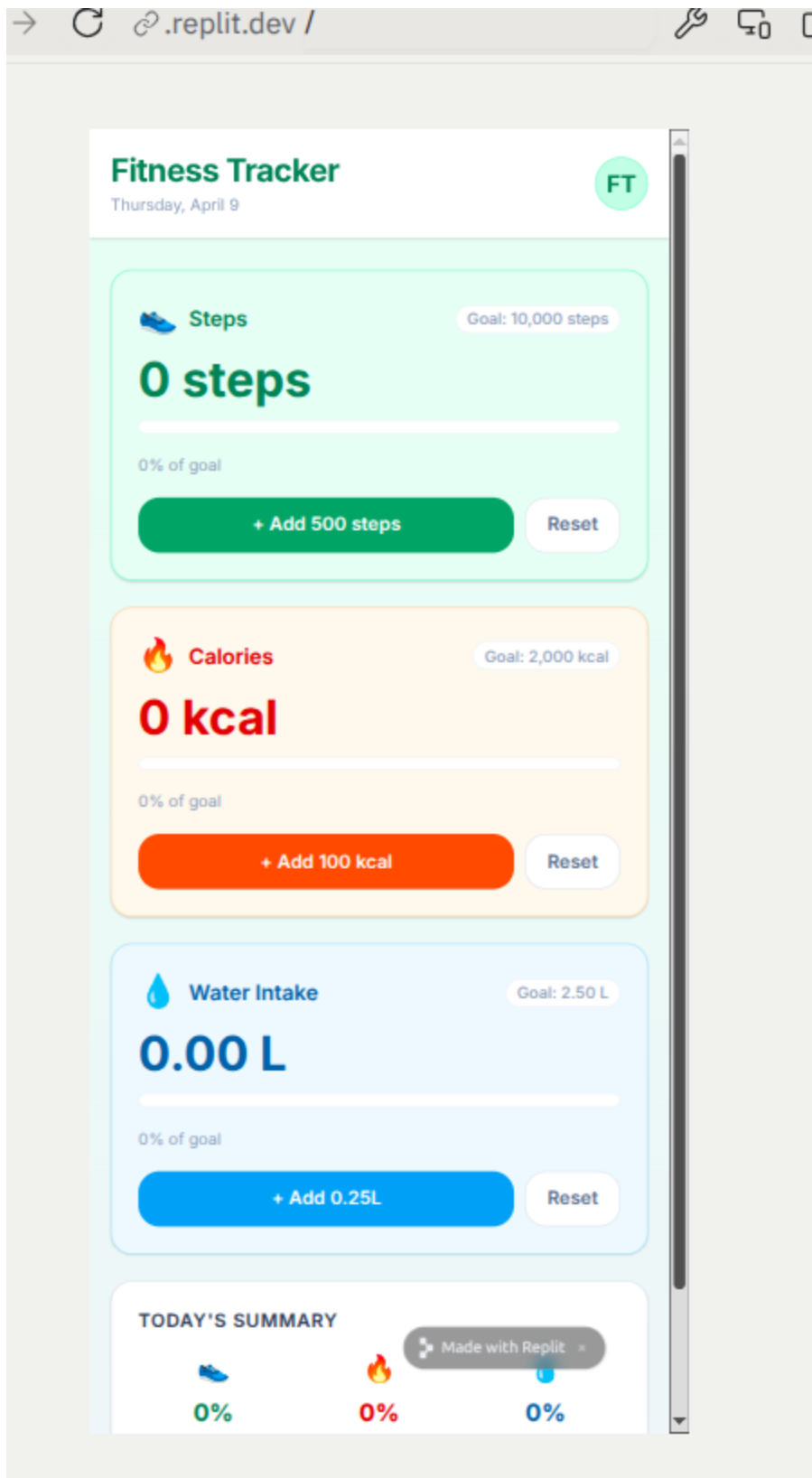
AI-generated layout with clearly visible sections

Thursday, April 9





Responsive design for mobile view



Task 2 – User Input & State Handling

Task:

Develop interactive features for user input.

Prompt used:

Enhance the fitness tracker app by:

- Adding input fields for steps, calories, and water intake
- Add buttons to update each value
- Use React useState to manage values
- Update values dynamically on screen
- Add validation to allow only numeric input

Output:

- >Input fields added
- >Buttons for updating values

Fitness Tracker

Thursday, April 9

FT

 **Steps** Goal: 10,000 steps

0 steps

0% of daily goal

+ Add Set

Reset

 **Calories** Goal: 2,000 kcal

0 kcal

0% of daily goal

+ Add Set

Reset

 **Water Intake** Goal: 2.50 L

0.00 L

0% of daily goal

+ Add Set

Reset

TODAY'S SUMMARY

**0%**
Steps
0 steps

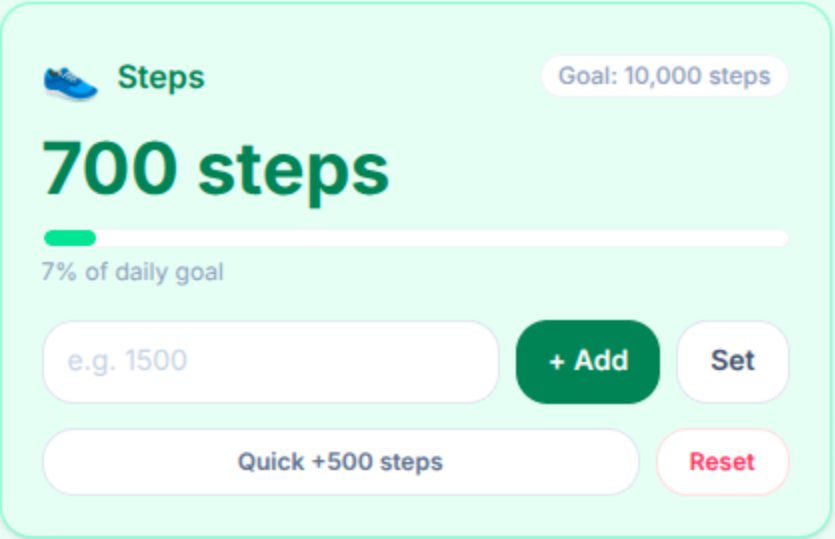
**0%**
Calories
0 kcal

**0%**
Water Intake
0.00 L

Type a value and tap + Add to log, or Set to replace the current value.

Values update instantly after clicking button

Thursday, April 9



Steps

Goal: 10,000 steps

700 steps

7% of daily goal

e.g. 1500

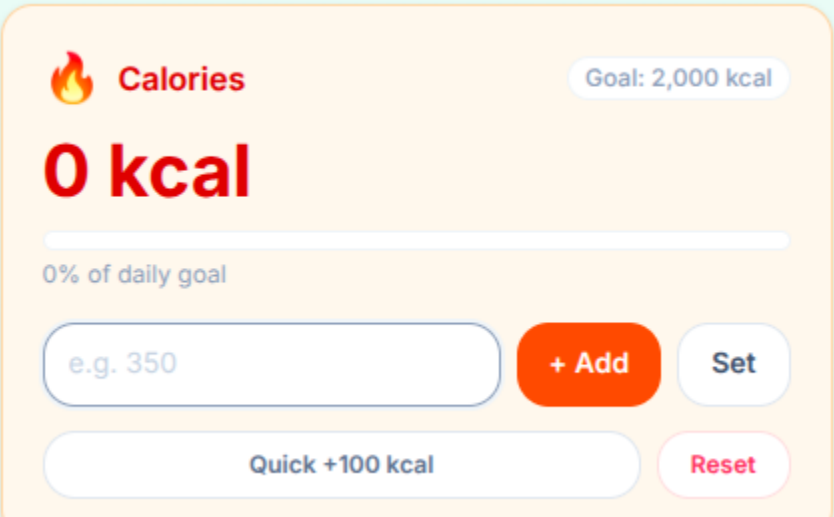
+ Add

Set

Quick +500 steps

Reset

Invalid input (text/symbols) is rejected



Calories

Goal: 2,000 kcal

0 kcal

0% of daily goal

e.g. 350

+ Add

Set

Quick +100 kcal

Reset

->State is maintained using React

Task 3 – Multi-Screen Navigation

Task:

Add navigation between multiple screens.

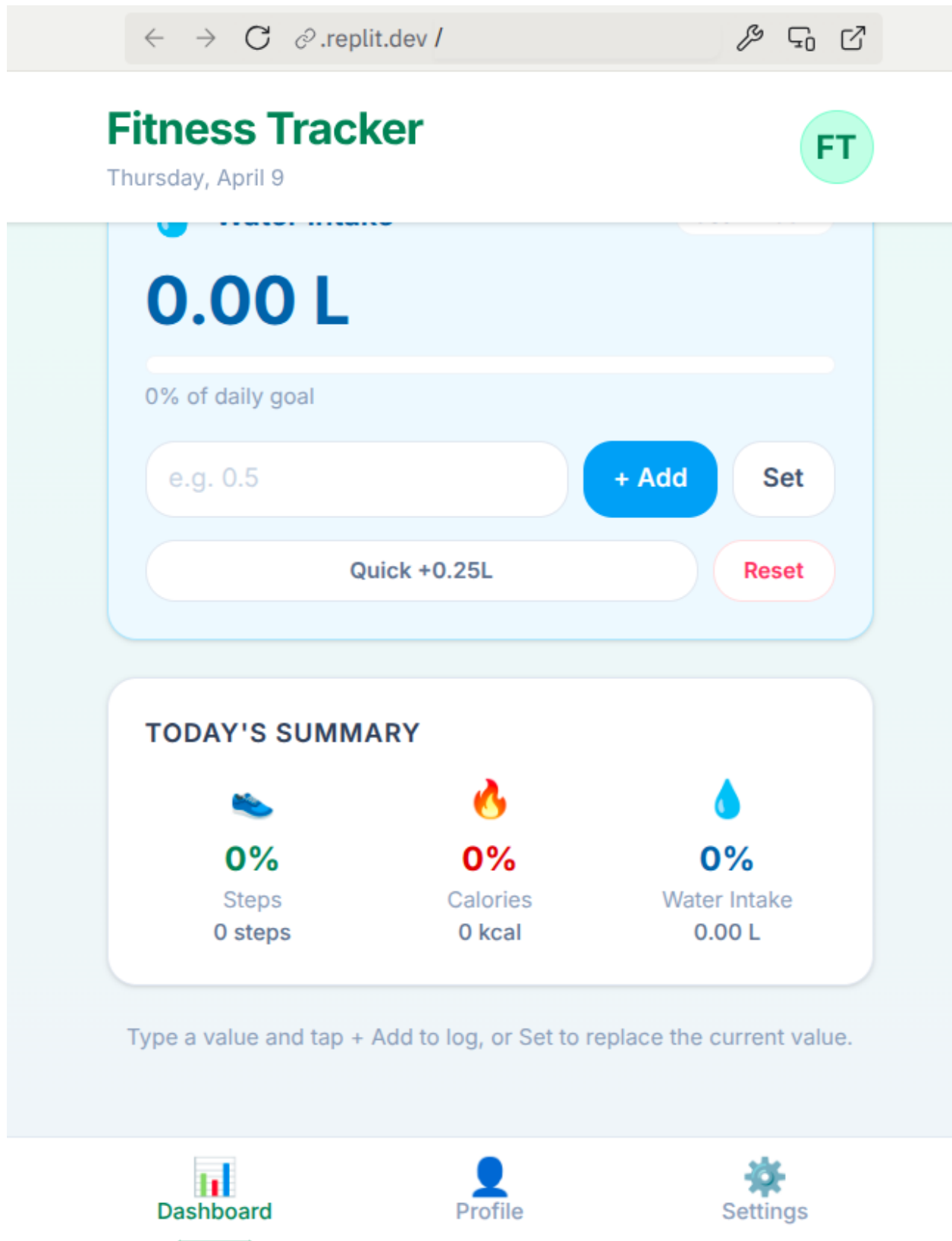
Prompt used:

Add navigation to the app:

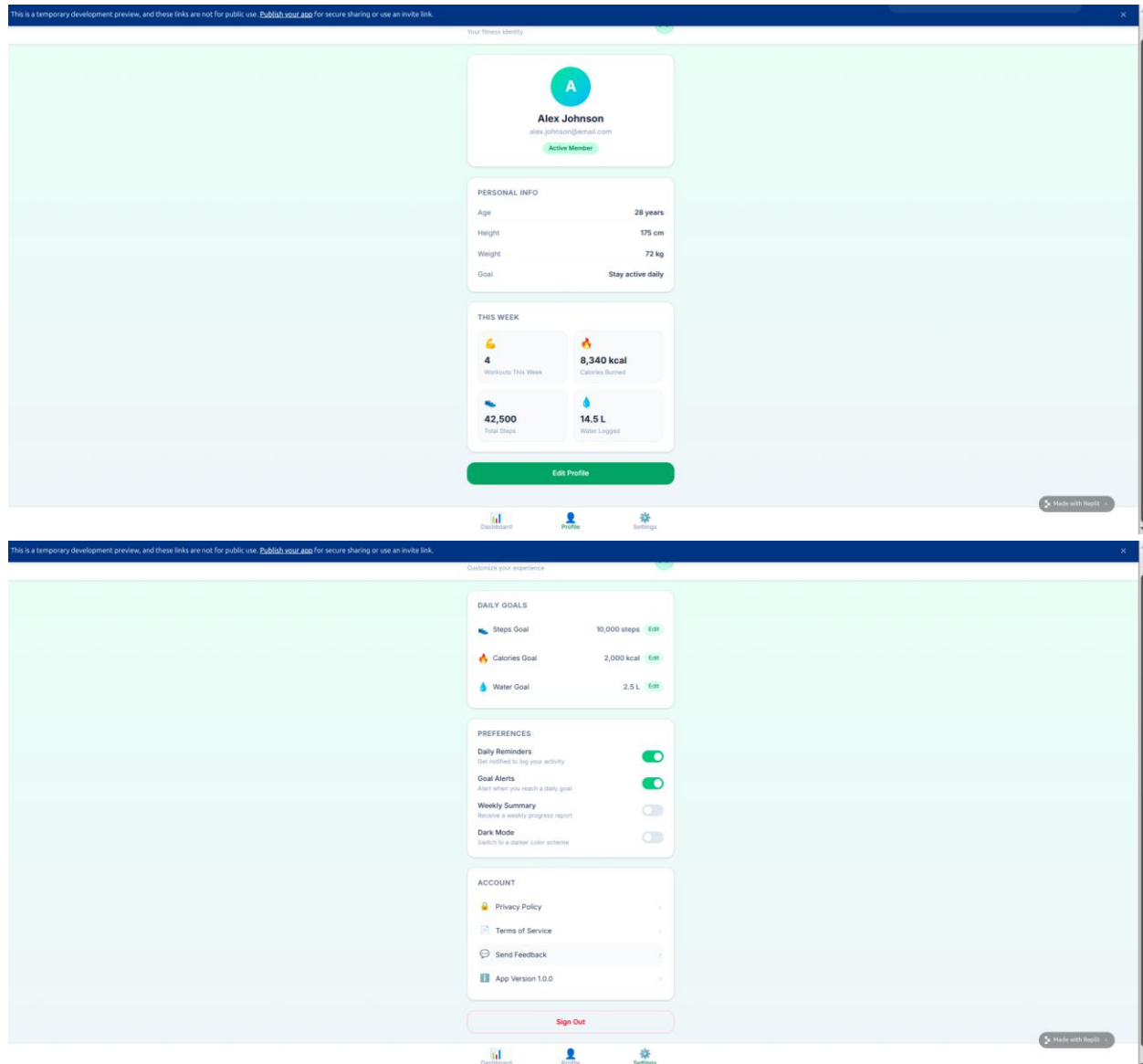
- Create 3 screens: Dashboard, Profile, Settings
- Use buttons or tabs for navigation
- Ensure switching screens does not reload the page
- Keep UI consistent across all screens
- Add simple placeholder content in Profile and Settings

Output:

Working navigation system



Multiple screens accessible without reload
Clean UI transitions



Task 4 – AI-Assisted UI Enhancement & Error Fixing

Task:

Improve UI and fix layout issues.

Prompt used:

Improve the UI and fix responsiveness issues:

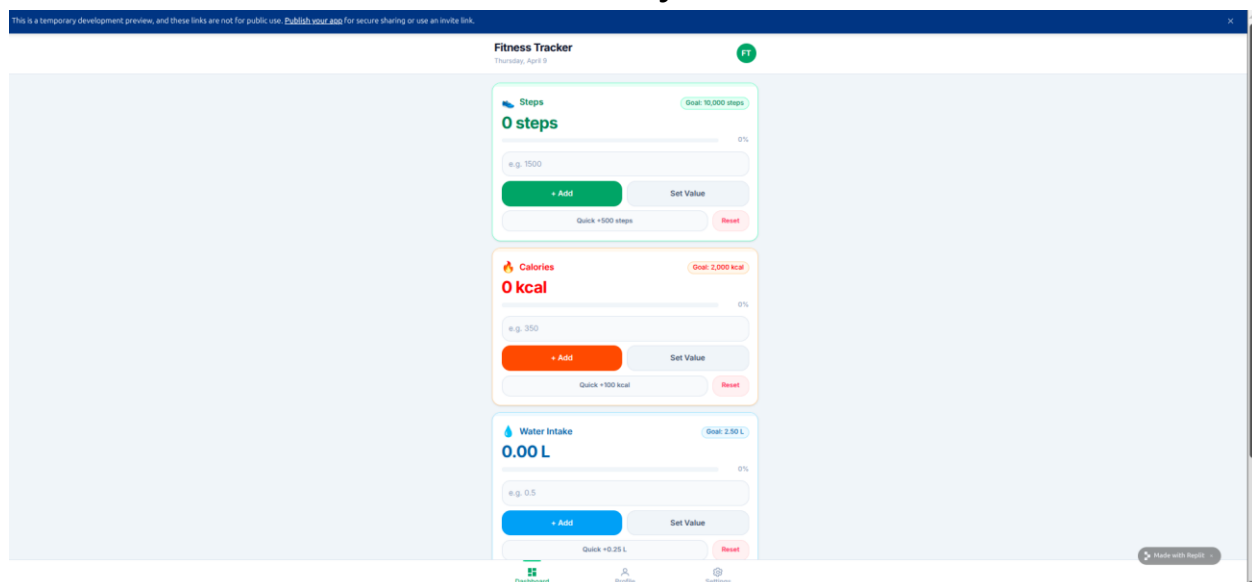
- Identify alignment problems on small screens
- Improve spacing, font sizes, and layout
- Make buttons and inputs mobile-friendly
- Use better colors and styling
- Ensure all elements align properly on different screen sizes

Output:

Improved UI design

Proper alignment across screen sizes

Cleaner and more readable layout



Mobile view

Fitness Tracker

Thursday, April 9

FT



Steps

Goal: 10,000 steps

0 steps

0%

e.g. 1500

Set Value

Quick +500 steps

Reset



Calories

Goal: 2,000 kcal

0 kcal

0%

e.g. 350

Set Value

Quick +100 kcal

Reset



Water Intake

Goal: 2.50 L

0.00 L

0%

Made with Replit



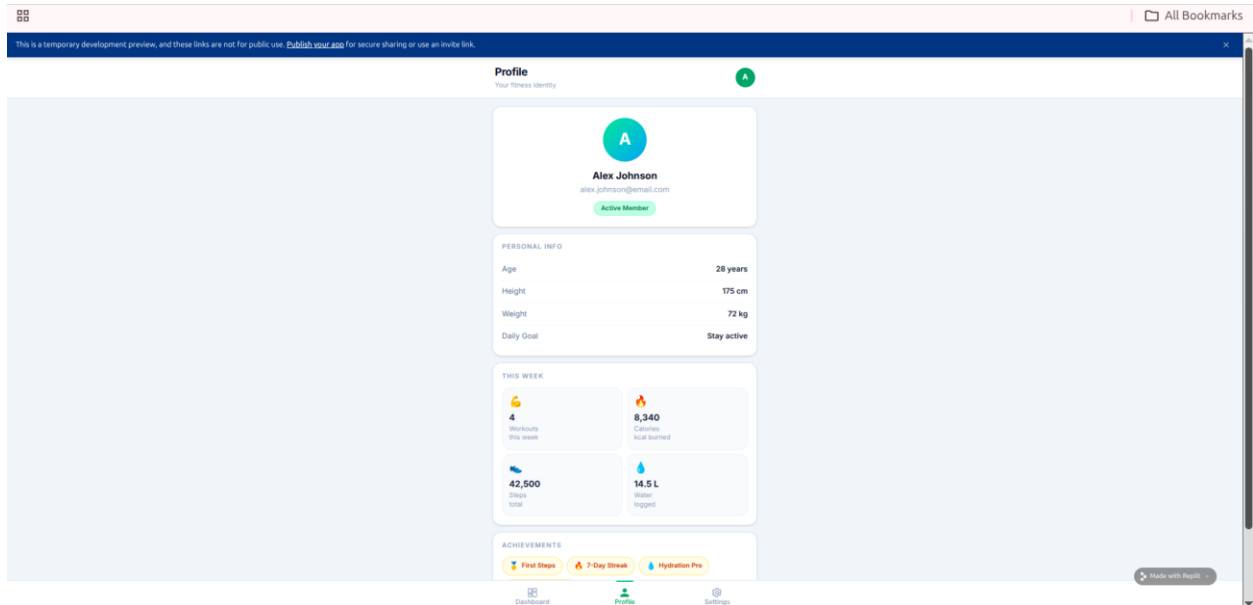
Dashboard



Profile



Settings



Task 5 – Sharing, Feedback & Iteration

Task:

Deploy and refine the app based on feedback.

Prompt used;

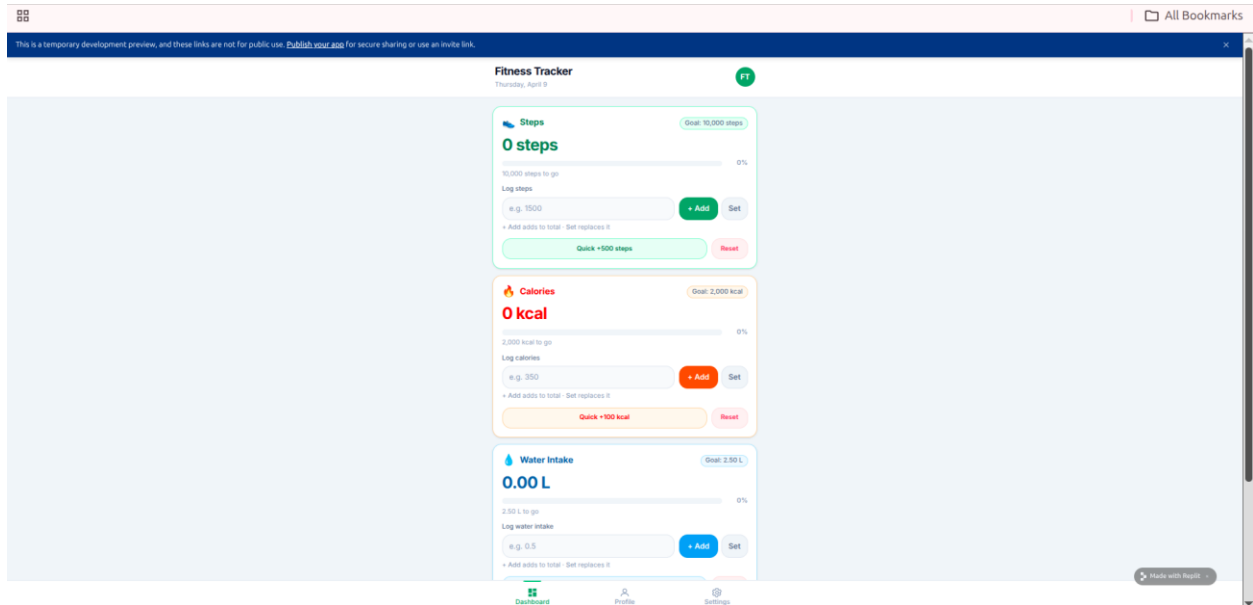
Suggest improvements based on user feedback:

- Improve UI clarity and usability
- Fix any bugs in input handling or navigation
- Enhance user experience with small changes
- Keep the app simple but more user-friendly

App link:

<https://fit-tracker-ui--shashidhar202.replit.app>

Improved version



Documented Feedback

User 1 Feedback

UI looks slightly crowded on mobile screens

Buttons are too close to each other

Input field not clearly labeled

User 2 Feedback

Confusion about which button updates which value

Navigation buttons not visually distinct

Water intake unit (L) not clearly understood

Issues Identified (from feedback)

Poor spacing between UI elements

Lack of labels for inputs

Buttons unclear in purpose

Weak visual hierarchy

Minor responsiveness issues

Improvements Implemented

UI Improvements

Added proper spacing between sections and buttons

Increased padding for better touch interaction

Improved font size for readability

Input Enhancements

Added labels:

Enter Steps

Enter Calories

Enter Water Intake (Liters)

Button Clarity

Renamed buttons:

Update Steps

Update Calories

Update Water

Navigation Enhancement

Styled navigation buttons (highlight active tab)

Added better alignment and spacing

Responsiveness Fix

Converted layout to column format for mobile

Ensured all elements fit within screen width

Improved Version Summary

Cleaner and more organized UI

Better readability on small screens

Clear input instructions

Improved user interaction

More intuitive navigation

Fully responsive design

Final output:

The application was deployed using Replit and shared with peers for feedback.

Based on user feedback, several improvements were implemented including better spacing, clearer input labels, improved button naming, and enhanced responsiveness.

The updated version provides a cleaner UI, improved usability, and a more intuitive user experience.